

Practical Week -1

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

1. Write a C code to create a process using fork() system call.
2. Read the user input as shell command.
3. Use a system call exec() to execute the shell command.
4. Use a wait() system call for waiting a parent process.
5. Display the PIDs of parent and child processes.

Example Output:

Enter a Linux command: ls

Parent Process

Child Process

Parent PID : 67

Child PID : 68

Child PID : 68

Parent PID : 67

Executing command: ls

Child process completed.

Sol:

```
satwika@SATWIK: ~/command_executor.c

GNU nano 8.7.1 command_executor.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    else if (pid == 0)
    {
        // Child process

        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);
    }

    // Parent process
    wait(NULL);
    printf("Child process completed.\n");
}
```

```
satwika@SATWIK: ~/command_executor.c

GNU nano 8.7.1 command_executor.c *
}
```

```

else if (pid == 0)
{
    // Child process

    printf("Child Process\n");
    printf("Child PID : %d\n", getpid());
    printf("Parent PID : %d\n", getppid());

    printf("Executing command: %s\n", command);

    execlp(command, command, NULL);

    perror("exec failed");
    exit(1);
}

else
{
    // Parent process

    printf("Parent Process\n");
    printf("Parent PID : %d\n", getpid());

    wait(NULL);

    printf("Child process completed.\n");
}

return 0;
}

```

^G Help

^K Cut

^R Read File

^J Justify

^O Write Out

^T Execute

^N Replace

^F Where Is

^X Exit

^U Paste

satwika@SATWIKA: ~

```

satwika@SATWIKA:~$ ls
File1.txt      ecec_demo.c      fork_exec.c      task2.c
File2.txt      exec_demo        namedirectory    task3
OSSP           exec_demo.c      skill4           task3.c
OS_LAB         filename         task1
command_executor.c.save filename.c        task1.c
ecec_demo      fork_exec        task2

satwika@SATWIKA:~$ nano command_executor.c
satwika@SATWIKA:~$ gcc command_executor.c -o command_executor
satwika@SATWIKA:~$ ./command_executor
Enter a Linux command: ls
Parent Process
Parent PID : 550
Child Process
Child PID : 551
Parent PID : 550
Executing command: ls
File1.txt      ecec_demo.c      skill4
File2.txt      exec_demo        task1
OSSP           exec_demo.c      task1.c
OS_LAB         filename         task2
command_executor filename.c        task2.c
command_executor.c fork_exec        task3
command_executor.c.save fork_exec.c      task3.c
ecec_demo      namedirectory
Child process completed.
satwika@SATWIKA:~$ ./command_executor
Enter a Linux command: pwd
Parent Process
Parent PID : 617
Child Process
Child PID : 620
Parent PID : 617
Executing command: pwd
/home/satwika
Child process completed.
satwika@SATWIKA:~$

```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Sol:

Command	What it shows

uname	Linux/system information
uname -a	Detailed kernel/system information
lscpu	CPU architecture and processor information
ps	Current processes
ps aux	Detailed process information
top	Real-time process and resource usage

satwika@SATWIKA: ~

```
satwika@SATWIKA:~$ uname
Linux
satwika@SATWIKA:~$ uname -a
Linux SATWIKA 6.18.33.2-microsoft-standard-WSL2 #1 SMP PRE
EMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Li
satwika@SATWIKA:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits vi
Byte Order:                   Little Endian
CPU(s):                       12
On-line CPU(s) list:         0-11
Vendor ID:                    GenuineIntel
Model name:                   13th Gen Intel(R) Core(TM) i
CPU family:                   6
Model:                        186
Thread(s) per core:          2
Core(s) per socket:          6
Socket(s):                    1
Stepping:                     3
BogoMIPS:                     4991.99
Flags:                        fpu vme de pse tsc msr pae m
                             apic sep mtrr pge mca cmov p
                             6 clflush mmx fxsr sse sse2
                             yscall nx pdpe1gb rdtscp lm
                             t_tsc rep_good nopl xtopolog
                             eliable nonstop_tsc cpuid ts
                             _freq pni pclmulqdq vmx ssse
                             x16 pcid sse4_1 sse4_2 x2api
                             popcnt tsc_deadline_timer a
                             e avx f16c rdrand hypervisor
                             m abm 3dnowprefetch ssbd ibr
                             stibp ibrs_enhanced tpr_shad
                             vpid ept_ad fsgsbase tsc_adj
                             1 avx2 smep bmi2 erms invpci
                             d adx smap clflushopt clwb s
                             saveopt xsavec xgetbv1 xsave
```

satwika@SATWIKA: ~

```
satwika@SATWIKA:~$ ps
  PID TTY          TIME CMD
  319 pts/0    00:00:00 bash
  733 pts/0    00:00:00 ps
satwika@SATWIKA:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TI
ME COMMAND
satwika    1  0.3  0.3  24328 15372 ?        Ss   05:20   0:
03 /sbin/init
satwika    2  0.0  0.0   3180  2208 hvc0    Sl+  05:20   0:
00 /init
satwika    7  0.0  0.0   3180  2048 hvc0    Sl+  05:20   0:
00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-f
satwika   53  0.0  0.4  50396 16804 ?        S<s  05:20   0:
00 /usr/lib/systemd/systemd-journald
systemd+   70  0.0  0.3  22416 14468 ?        Ss   05:20   0:
00 /usr/lib/systemd/systemd-resolved
satwika   92  0.1  0.3  35140 12256 ?        Ss   05:20   0:
01 /usr/lib/systemd/systemd-udev
satwika   153  0.0  0.0   2888  1948 ?        Ss   05:20   0:
00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
satwika   154  0.0  0.0   4472  3100 ?        Ss   05:20   0:
00 /usr/sbin/cron -f -P
message+  155  0.0  0.1   8820  5316 ?        Ss   05:20   0:
00 @dbus-daemon --system --address=systemd: --nofork --nopidfile
satwika   159  0.0  0.7  44096 29712 ?        Ss   05:20   0:
00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-t
satwika   168  0.0  0.2  18436  9388 ?        Ss   05:20   0:
00 /usr/lib/systemd/systemd-logind
syslog    175  0.0  0.1 220548  5568 ?        Ssl  05:20   0:
00 /usr/sbin/rsyslogd -n -iNONE
```

```
satwika      216  0.0  0.0   5492  2660 tty1      Ss+  05:20  0:
00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:
satwika      225  0.0  0.8 123460 32772 ?        Ssl  05:20  0:
00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-up
_chrony      226  0.0  0.2  23588 11084 ?        S    05:20  0:
00 /usr/sbin/chronyd -n -F 1 -x
```

satwika@SATWIKA: ~

```
satwika@SATWIKA:~$ top
top - 05:43:42 up 2:51, 1 user, load average: 0.09, 0.04, 0.
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 z
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi,
MiB Mem : 3758.6 total, 3048.4 free, 473.1 used, 312.
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used. 3285.

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM
    1 satwika    20   0   24328  15372 11568 S   0.0   0.4
    2 satwika    20   0    3180   2208  2072 S   0.0   0.1
    7 satwika    20   0    3180   2048  1940 S   0.0   0.1
   53 satwika    19  -1   50396  16804 15500 S   0.0   0.4
   70 systemd+   20   0   22416 14468 11972 S   0.0   0.4
   92 satwika    20   0   35140  12256  9184 S   0.0   0.3
  153 satwika    20   0   2888   1948  1820 S   0.0   0.1
  154 satwika    20   0    4472   3100  2852 S   0.0   0.1
  155 message+   20   0    8820   5316  4636 S   0.0   0.1
  159 satwika    20   0   44096  29712 14576 S   0.0   0.8
  168 satwika    20   0   18436   9388  8144 S   0.0   0.2
  175 syslog     20   0 220548   5568  4648 S   0.0   0.1
  216 satwika    20   0    5492   2660  2476 S   0.0   0.1
  225 satwika    20   0 123460  32772 18164 S   0.0   0.9
  226 _chrony     20   0   23588  11084  7128 S   0.0   0.3
  237 _chrony     20   0   12096   2452  1808 S   0.0   0.1
  316 satwika    20   0    3184   1112   980 S   0.0   0.0
  317 satwika    20   0    3200   1248  1108 S   0.0   0.0
  319 satwika    20   0    6268   5532  3864 S   0.0   0.1
  322 satwika    20   0    8648   5544  4700 S   0.0   0.1
  385 satwika    20   0   22340  13604 10976 S   0.0   0.4
  387 satwika    20   0   22920   4092  2088 S   0.0   0.1
  412 satwika    20   0    6136   5424  3864 S   0.0   0.1
  755 satwika    20   0    8164   6064  3948 R   0.0   0.2
```